



## Practice Assignment 11 - Not Graded



This assignment will not be graded and is only for practice.

1 point

Let  $f(x, y)$  be defined in a neighbourhood of  $(a, b)$  and  $D(x, y) = f_{xx}f_{yy} - f_{xy}^2$

$D(x, y) = f_{xx}f_{yy} - f_{xy}^2$ . Suppose  $(a, b)$  is a critical point of  $f$ . Which of the following option(s) is(are) true?

☐

If  $D(a, b) > 0$  and  $f_{xx}(a, b) > 0$ , then  $(a, b)$  is a point of local minimum.

☐

If  $D(a, b) < 0$ , then  $(a, b)$  is a point of local maximum.

☐

If  $D(a, b) = 0$ , then  $(a, b)$  may be a point of minimum or maximum or neither.

☐

If  $D(a, b) < 0$ , then  $(a, b)$  is a saddle point.

**No, the answer is incorrect.**

**Score: 0**

**Accepted Answers:**

If  $D(a, b) > 0$  and  $f_{xx}(a, b) > 0$ , then  $(a, b)$  is a point of local minimum.

If  $D(a, b) = 0$ , then  $(a, b)$  may be a point of minimum or maximum or neither.

If  $D(a, b) < 0$ , then  $(a, b)$  is a saddle point.

1 point

Choose the correct statement(s) about  $f(x, y) = x^4 + y^4$ , where

$D(x, y) = f_{xx}f_{yy} - f_{xy}^2$

☐

$(0, 0)$  is the only critical point.

☐

$D(0, 0) > 0$  and  $f_{xx}(0, 0) < 0$  and so  $(0, 0)$  is a point of local maximum.

☐

$D(0, 0) = 0$ .

☐

$(0, 0)$  is a point of local minimum.

**No, the answer is incorrect.**

**Score: 0**

**Accepted Answers:**

$(0, 0)$  is the only critical point.

$D(0, 0) = 0$ .

$(0, 0)$  is a point of local minimum.

1 point

Let  $f(x, y) = (9x^2 - 1)(1 + 4y)$  be defined on the box  $-2 \leq x \leq 3$  and  $-1 \leq y \leq 4$ . Then which of the following options is (are) true?

☐

$(0, 1)$  is a point of local minimum.

☐

$(3, 4)$  is a point of absolute maximum.

☐

$-105$  is the absolute minimum.



☐

$(-2, -1)$  is a point of absolute minimum.

**No, the answer is incorrect.**

**Score: 0**

**Accepted Answers:**

$(3, 4)$  is a point of absolute maximum.

1 point

Suppose  $z = f(x) + yg(x)$  for some polynomial functions  $f$  and  $g$ . Which of the following options is true?

☐

$z_{yy} = 0$

☐

$z_{yy}$  is a non-constant function of  $x$ .

☐

$z_{yy}$  is a non-constant function of  $y$ .

☐

$z_{yy} = 1$

**No, the answer is incorrect.**

**Score: 0**

**Accepted Answers:**

$z_{yy} = 0$

1 point

Define a function

$$f(x, y) = \begin{cases} \frac{xy(x^2 - y^2)}{x^2 + y^2} & \text{if } (x, y) \neq (0, 0) \\ 0 & \text{if } (x, y) = (0, 0) \end{cases}$$

$(0, 0)$  if  $(x, y) = (0, 0)$

Choose the set of correct options.

☐

$f_{xy}(0, 0) = -1$

☐

$f_{yx}(0, 0) = 1$

☐

$f_{xy}(0, 0) = f_{yx}(0, 0)$

☐

$f_x(0, 0) = f_y(0, 0)$

**No, the answer is incorrect.**

**Score: 0**

**Accepted Answers:**

$f_{xy}(0, 0) = -1$

$f_{yx}(0, 0) = 1$

$f_x(0, 0) = f_y(0, 0)$

1 point

Consider a function  $f(x, y, z) = x^3y + yz$ . If  $A$  is the Hessian matrix of  $f(x, y, z)$  at  $(1, 1, 0)$ , then which of the following option(s) is(are) true?

☐

$\det(A) = 0$

☐

$$\text{Rank}(A) = 2 \text{ Rank}(A) = 2$$



$$\text{Nullity}(A) = 1 \text{ Nullity}(A) = 1$$



$$\det(A) = -6 \det(A) = -6$$



$$\text{Rank}(A) = 3 \text{ Rank}(A) = 3$$

**No, the answer is incorrect.**

**Score: 0**

**Accepted Answers:**

$$\det(A) = -6 \det(A) = -6$$

$$\text{Rank}(A) = 3 \text{ Rank}(A) = 3$$

1 point

Let  $f: \mathbb{R}^6 \rightarrow \mathbb{R}$  be a polynomial function. Which of the following options is true?

☐ Hessian matrix of the function is a symmetric matrix of order 6.



All possible mixed partial derivatives of second order have the same value at a point of  $\mathbb{R}^6$

☐ Hessian matrix of the function is a symmetric matrix of order 5.

☐ Hessian matrix of the function is a symmetric matrix of order 4.

**No, the answer is incorrect.**

**Score: 0**

**Accepted Answers:**

Hessian matrix of the function is a symmetric matrix of order 6.

Let  $S$  be a rectangular box. Let  $x, y, z$  be the length of the sides of the box and volume of the box  $V = xyz$ . If  $d$  is the length of the diagonal of the rectangular box, then

$$d = \sqrt{x^2 + y^2 + z^2} \quad d = \sqrt{x^2 + y^2 + z^2}$$



. Use this information to answer the questions 8 and 9.

1 point

If the diagonal is of length 2 units and the volume of the box is the maximum possible, then which of the following option(s) is (are) true?



$S$  is a cube.

☐ All sides are of different length.



$$x = \frac{2}{\sqrt{3}} \quad x = \frac{2}{\sqrt{3}}$$



$$y = \sqrt{3} \quad y = \sqrt{3}$$



$$y = \sqrt{3} \quad y = \sqrt{3}$$



☐ None of the above.

**No, the answer is incorrect.**

**Score: 0**

**Accepted Answers:**

$S$  is a cube.

$$x = \frac{2}{\sqrt{3}} \quad x = \frac{2}{\sqrt{3}}$$

√

2

1 point

If the length of the diagonal of the box is 3 units, then which of the followings is the maximum possible volume of the box?

☐ $\sqrt{3}$  3

√

cube units.

☐ $3\sqrt{3}$  3

√

cube units.

☐ 1 cube units.☐

33 cube units

**No, the answer is incorrect.**

**Score: 0**

**Accepted Answers:**

 $3\sqrt{3}$  3

√

cube units.

1 point

Consider the following function

$$f(x, y, z) = x^2 y^2 + x z^2 \quad f(x, y, z) = x^2 y^2 + x z^2$$

Which of the following option(s) is (are) true?

☐

$(0, y, 0)$ , where  $y \in \mathbb{R}$ , where  $y \in \mathbb{R}$  are critical points.

☐

$(0, 0, z)$ , where  $z \in \mathbb{R}$ , where  $z \in \mathbb{R}$  are critical points.

☐

$(1, y, 0)$  where  $y \in \mathbb{R}$ , where  $y \in \mathbb{R}$  are critical points.

☐

$(0, y, 1)$ , where  $y \in \mathbb{R}$ , where  $y \in \mathbb{R}$  are critical points.

☐

$(x, 0, 0)$ , where  $x \in \mathbb{R}$ , where  $x \in \mathbb{R}$  are critical points.

**No, the answer is incorrect.**

**Score: 0**

**Accepted Answers:**

$(0, y, 0)$ , where  $y \in \mathbb{R}$ , where  $y \in \mathbb{R}$  are critical points.

$(x, 0, 0)$ , where  $x \in \mathbb{R}$ , where  $x \in \mathbb{R}$  are critical points.

[Check Answers](#)

Your score is: 0/10